

Farm Animal Demo Program

Java Lab Record

1. Aim

To develop a Java program to demonstrate classes, objects, constructors, and methods by storing and displaying the details of different farm animals.

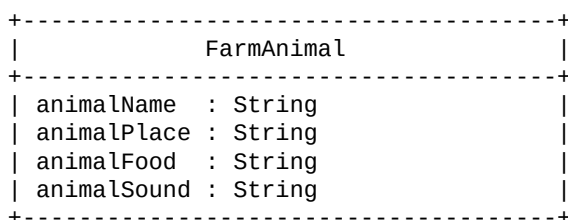
2. Step-by-Step Procedure

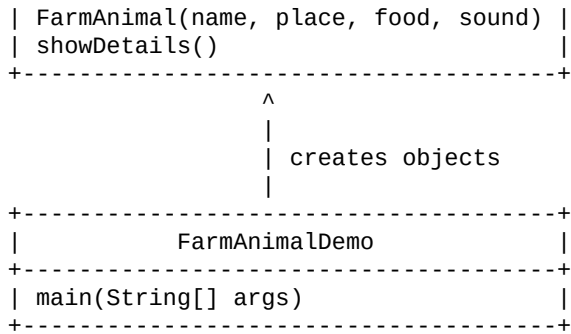
1. Create a class named FarmAnimal.
2. Declare animalName, animalPlace, animalFood, and animalSound as data members.
3. Create a parameterized constructor to initialize the animal details.
4. Create a method named showDetails() to display all the details of an animal.
5. Create the FarmAnimalDemo class containing the main() method.
6. Create an object named cow with the details of a cow.
7. Create an object named pig with the details of a pig.
8. Create an object named horse with the details of a horse.
9. Call the showDetails() method for each object.
10. Display the details of all the farm animals.
11. Stop the program.

3. Algorithm

1. Start.
2. Create the FarmAnimal class.
3. Declare variables for name, place, food, and sound.
4. Create a constructor to initialize the animal details.
5. Create the showDetails() method to display the details.
6. Create objects for Cow, Pig, and Horse.
7. Call showDetails() for the Cow object.
8. Call showDetails() for the Pig object.
9. Call showDetails() for the Horse object.
10. Stop.

4. UML Class Diagram





5. Code

```

class FarmAnimal {
    String animalName;
    String animalPlace;
    String animalFood;
    String animalSound;

    FarmAnimal(String animalName,String animalPlace,String animalFood,String animalSound) {
        this.animalName=animalName;
        this.animalPlace=animalPlace;
        this.animalFood=animalFood;
        this.animalSound=animalSound;
    }

    void showDetails() {
        System.out.println("Animal Name : "+animalName);
        System.out.println("Lives At    : "+animalPlace);
        System.out.println("Food       : "+animalFood);
        System.out.println("Sound      : "+animalSound);
        System.out.println();
    }
}

public class FarmAnimalDemo {
    public static void main(String[] args) {
        FarmAnimal cow=new FarmAnimal("Cow","Cowshed","Grass","Moo");

        FarmAnimal pig=new FarmAnimal("Pig","Pigsty","Vegetables","Oink");

        FarmAnimal horse=new FarmAnimal("Horse","Stable","Grass","Neigh");

        cow.showDetails();
        pig.showDetails();
        horse.showDetails();
    }
}

```

6. Sample Output

The following output is obtained by executing the given FarmAnimalDemo program.

Animal Name : Cow

Lives At : Cowshed

Food : Grass

Sound : Moo

Animal Name : Pig

Lives At : Pigsty

Food : Vegetables

Sound : Oink

Animal Name : Horse

Lives At : Stable

Food : Grass

Sound : Neigh